FerroSelect

An AI model that predicts ferroptosis sensitivity in triple-negative breast cancer, tested blind against cell-culture experiments using low-cost drugs

Undergraduate graduation project proposal, 2026 to 2027

Discipline	Medical biotechnology, using machine learning as the enabling tool
Institution	Department of Biochemistry, Faculty of Science, Alexandria University
Team	5 final-year Biochemistry students. Two work on the computational half, three at the bench
Supervisor	[Name, title, department]
Duration	9 months, November 2026 to July 2027
Budget requested	100,000 EGP, which is 20,000 EGP per student
Where it ends up	TRL 2 at the start, TRL 4 at the end: a prediction validated in a laboratory model
Keywords	triple-negative breast cancer, ferroptosis, GPX4, System xc-, machine learning, drug repurposing, blinded validation, sulfasalazine

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1. Summary

Breast cancer is the most common cancer among Egyptian women. It makes up about a third of all female cancers in the national registry [7], and roughly 13% of those tumours are triple-negative [8], which is at the high end of what other countries report. A triple-negative tumour carries no oestrogen receptor, no progesterone receptor and no HER2 amplification, so hormone therapy and HER2 antibodies do nothing for it. Chemotherapy is what is left, and relapse is common [6].

Ferroptosis is a different way to kill a cell. It runs on iron and on lipid peroxidation rather than on the apoptosis machinery that resistant cells often learn to switch off [1,2]. Cancer cells sitting in a drug-tolerant, mesenchymal state turn out to lean hard on one enzyme, GPX4, to survive it [4,5]. Several drugs that push cells toward ferroptosis are already sold in Egypt as cheap generics.

So which tumours would respond? The published models answer this in a way that cannot really be checked. They score each sample for ferroptosis sensitivity using a gene set, then train a classifier to predict that score from the same expression matrix the score was calculated from. The classifier recovers the formula. It never meets a cell. Papers built this way report accuracies above 0.9, and validating on a second cohort does not rescue them, because recomputing the same formula somewhere else will always agree with itself.

We do three things differently. We train on drug response that somebody actually measured, meaning the IC50 and AUC values in the large public cell-line screens [13,14,15], matched to expression profiles from CCLE [16]. We then freeze the model, write down what it predicts, and lodge that prediction with our supervisor before anyone treats a single well. The three students running the cell work never see the ranking until their data is finished and locked. At the end we check that the death was really ferroptosis, using rescue agents that should reverse it and a caspase inhibitor that should not.

What comes out is a number we can defend: how often the model was right on cells it had never seen, scored by people who did not know what it had guessed. Alongside that we expect to leave a drug combination that costs very little, a small web tool anyone can use, and five graduates who have done both halves of the work.

2. Background

2.1 The clinical problem

Triple-negative breast cancer is defined by what it lacks, and what it lacks is every target that has made the other breast cancer subtypes more survivable. It is not even one disease. Lehmann and colleagues split it by expression into subtypes with different vulnerabilities, one of them mesenchymal [6]. In Egypt the picture is worse than the global average, because patients arrive younger and at a later stage [8]. Nothing in routine practice tells an oncologist which non-apoptotic death pathway a particular tumour might be open to.

2.2 How ferroptosis works

Ferroptosis is iron-dependent death that happens when lipid hydroperoxides build up in membranes faster than the cell can clear them [1,2]. Two points in that clearance system matter for this project.

The first is the System xc- antiporter, SLC7A11, which trades glutamate out for cystine in. Cystine becomes cysteine, cysteine feeds glutathione synthesis, and glutathione is the cofactor GPX4 needs to do its job [3]. Starve the cell of cystine and the whole chain fails.

The second is GPX4 itself, the enzyme that reduces lipid hydroperoxides to harmless alcohols. Inhibit it directly and you reach the same endpoint by a shorter route [3].

That gives two classes of drug. Type I inducers such as sulfasalazine and erastin block the transporter. Type II inducers such as RSL3 bind GPX4 covalently. There is also a slower, indirect route. GPX4 is a selenoprotein, and making it requires isopentenyl pyrophosphate from the mevalonate pathway to modify the selenocysteine tRNA [18]. Statins block that pathway. They do not cause ferroptosis on their own, but they can make a cell easier to kill, which is why we only ever use one in combination.

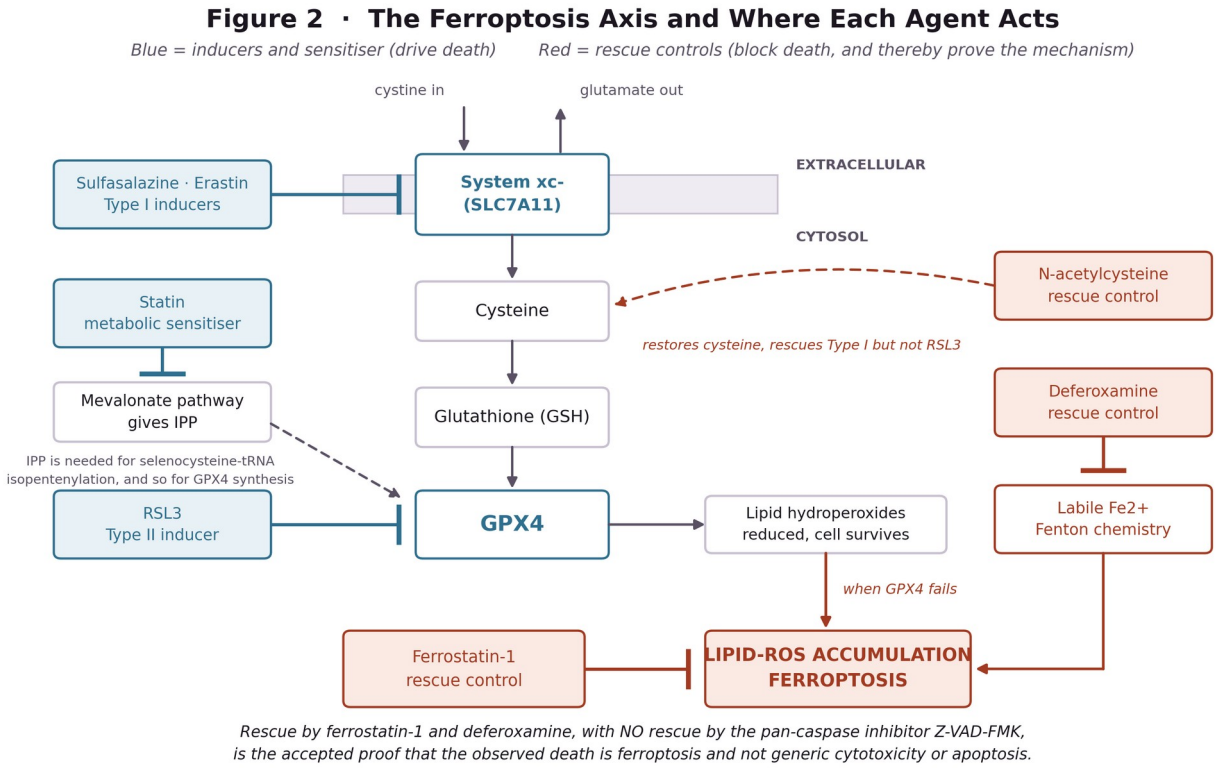


Figure 2. The ferroptosis axis, with the point of action of every agent in this project. Rescue by ferrostatin-1 and deferoxamine, together with no rescue by the caspase inhibitor Z-VAD-FMK, is the accepted way to show that what you are looking at is ferroptosis.

2.3 Why triple-negative breast cancer

Three findings pointed us here.

Drug-tolerant cancer cells in a mesenchymal state depend on GPX4 and die when it is inhibited [4,5]. Mesenchymal-like TNBC lines, MDA-MB-231 among them, sit in exactly that state.

The oestrogen receptor suppresses transferrin receptor expression, so ER-positive cells take up less iron and are harder to push into ferroptosis [9]. That is useful to us. It means MCF-7 should come out more resistant than MDA-MB-231 for a reason we can name in advance, instead of being an arbitrary comparison we added to fill a column.

And the size of the effect is already on record. Erastin kills MDA-MB-231 with an IC50 near 2.2 uM, and ferrostatin-1 reverses death caused by both erastin and RSL3 in that line [10]. We are not hoping something will happen. We know roughly what should happen, and that is what makes a blinded test worth running.

2.4 Why we prioritise computationally first

Bench time is the resource we do not have enough of. One full dose-response experiment in triplicate costs a fortnight of a student's life and a real fraction of the budget. Meanwhile the public screens already hold measured sensitivity for thousands of compounds across hundreds of cell lines, with matched transcriptomes sitting next to them [13,14,15,16]. Using that to decide what goes on a plate costs nothing. The model is not replacing the experiment. It is choosing which experiment to run.

2.5 The problem we want to measure

We ran into this while reading the ferroptosis biomarker literature. Much of it trains classifiers on single-sample enrichment scores of curated ferroptosis gene sets [17]. The label is computed from the same matrix that supplies the features, so the classifier is being asked to reproduce arithmetic. It usually can. The accuracy is real in a narrow sense and meaningless in every sense that matters to a patient.

Rather than complain about it, we decided to measure it. We build the circular model too, on identical data splits, and report the two side by side. We call this Aim 0. Of everything in this proposal, that comparison is the part most likely to be useful to somebody outside our department.

3. What is new here

Four things, listed in the order we would defend them.

	Claim	Why it holds up
1	The labels come from measured killing, not from a score we calculated ourselves	GDSC, CTRPv2 and PRISM record what a compound did to a cell line in somebody's laboratory [13,14,15]. Those numbers exist whether or not our features agree with them, so a model trained on them is capable of being wrong. That is the whole point
2	The prediction is sealed and the bench team is blinded	Written down, dated and signed before any treatment. The supervisor relabels the agents A to H, so the three students at the bench cannot know which compound they are dosing. The comparison at the end is a test rather than a fit, and a reviewer can check the timestamp
3	We measure how much the usual approach inflates its accuracy, instead of arguing about it	Both labelling schemes, identical splits, reported next to each other. We have not found this comparison published anywhere for ferroptosis, and it costs us about a week of compute to produce
4	Every actionable agent can be bought in Egypt	Sulfasalazine and simvastatin are registered generics here. We report cost per treatment course at local prices, so the result is something a health system could act on rather than read about

There is a fifth thing we would like to claim, and will only claim if it works. The model does not just rank cells. It also nominates a statin and inducer pairing that should turn a resistant line into a responsive one. Predicting that a sensitive cell dies is easy to believe. Designing a way to make a resistant cell die, and then having it work, is a much stronger result. We have costed that arm and put it in the plan, but we have not built the project on it.

4. Hypotheses and objectives

	Hypothesis
H1	Gene expression predicts measured sensitivity to ferroptosis inducers better than chance, and worse than the proxy-labelled literature claims
H2	The two TNBC lines are more sensitive to System xc- and GPX4 inhibition than MCF-7 is, for the oestrogen receptor and transferrin receptor reason described in section 2.3 [9]
H3	A sub-lethal dose of a statin lowers the IC50 of a ferroptosis inducer in at least one cell line

The general aim is to build a model that predicts ferroptosis sensitivity in TNBC from gene expression, train it on real drug response, and then test it blind against cells in our own laboratory using drugs that are affordable in Egypt.

	Objective	How we know it is done
1	Measure the accuracy inflation caused by ssGSEA-derived labels (Aim 0)	Both models reported on identical splits, with the gap stated as a number and a confidence interval
2	Assemble matched expression and measured sensitivity data for ferroptosis-relevant agents	A documented, reproducible dataset with the per-agent sample size written down honestly, including where it is small
3	Train the model and evaluate it without fooling ourselves	Nested cross-validation with whole cell lines held out. Performance compared against the published laboratory-versus-CTRPv2 agreement for GPX4 inhibitors [12], which tells us what good even looks like
4	Freeze the model and seal the prediction	A dated, signed document with the supervisor before the first plate is seeded
5	Test the prediction on three breast cell lines with two low-cost inducers	Eight-point dose-response, n = 3, IC50 for every cell line and agent pair
6	Show the death is ferroptosis, and report what the model actually scored	Rescue by ferrostatin-1 and deferoxamine, no rescue by Z-VAD-FMK, plus the predicted ranking compared against the measured one

5. How the project runs

Figure 1 · FerroSelect Closed-Loop Workflow

The computational prediction is locked and sealed before a single cell is treated

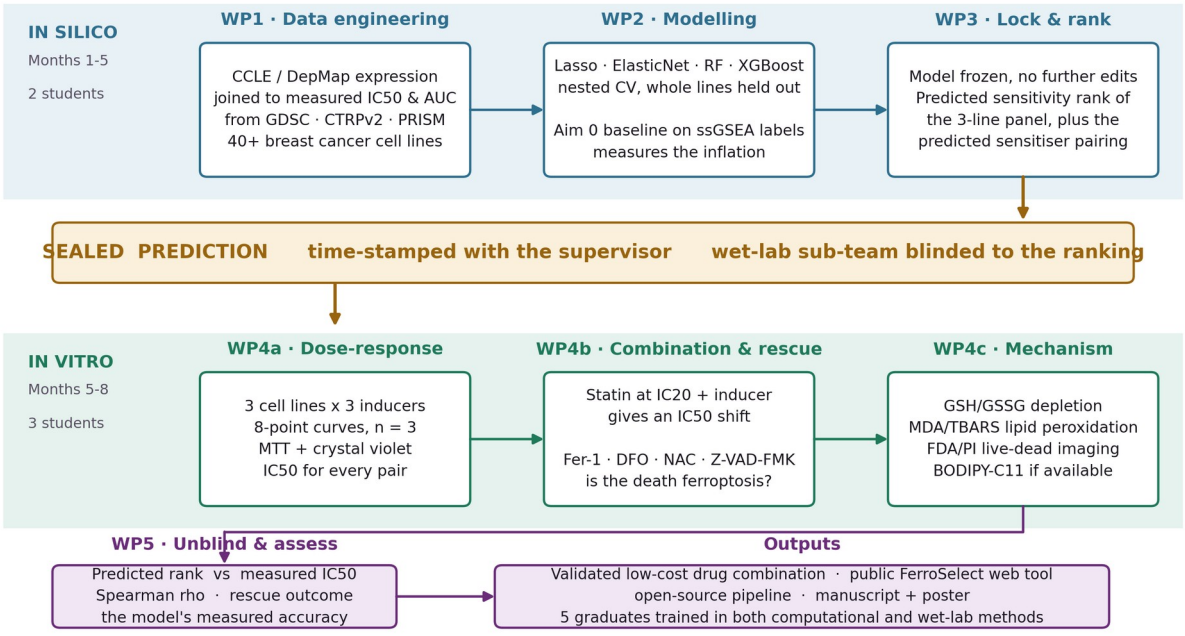


Figure 1. The two halves of the project, separated by a one-way gate. The prediction is locked and sealed before the bench work starts, and the bench team does not see the ranking until their assays are finished. That gate is what turns the final comparison into a test instead of a fit.

Phase	Months	What happens	Before we move on
WP1, data	1 to 2	Pull, harmonise and check the expression and sensitivity data	A documented dataset that somebody else could rebuild
WP2, modelling	2 to 4	Train the model and the Aim 0 baseline, then look at which genes drive it	Held-out performance and the inflation gap are both written down
WP3, lock	4 to 5	Freeze the model, generate the ranking, seal it, order reagents, revive the cells	Sealed prediction lodged, cell lines growing and mycoplasma-free
WP4, bench	5 to 8	Dose-response, rescue, combination and mechanism assays	Core arms complete at n = 3
WP5, unblind	8 to 9	Break the code, compare, deploy the tool, write it up	Outputs submitted

6. Experimental design

6.1 The computational half

Element	What we are doing
Unit of analysis	The cell line, because that is the level at which measured drug response actually exists. We use patient cohorts only to look at survival association, and we label that clearly as hypothesis-generating rather than as a result
Features	Expression from CCLE and DepMap [16], variance-filtered. A second, smaller model uses only a curated ferroptosis gene set [17] so that we can read what it is doing
Labels	Measured IC50 and AUC from GDSC [13], CTRPv2 [14] and PRISM [15]
Aim 0 labels	ssGSEA enrichment computed from the input expression matrix. Circular on purpose. We keep it so we can measure the inflation rather than assert it
Training strategy	Train across all tissues and hold the breast lines out. Breast-specific sample sizes are small for some of these compounds, and we would rather say so than pretend otherwise
Algorithms	Lasso and ridge regression, which we can interpret, and a random forest for the non-linear case. We are not using deep learning. There are a few hundred cell lines, not a few million, and a neural network here would be harder to defend and no more accurate
Validation	Nested cross-validation with whole cell lines held out. Screen identity goes in as a covariate, and we check per screen that no single dataset is driving the answer
Benchmark	The published agreement between laboratory IC50 and CTRPv2 AUC for GPX4 inhibitors is around 0.83 for RSL3, 0.70 for ML210 and 0.79 for ML162 [12]. That is roughly the ceiling. Anything we report above it should be treated as a bug rather than a triumph
Software	Python with pandas, scikit-learn, SciPy, statsmodels and SHAP. Roughly 200 lines of analysis code, kept in a public Git repository

6.2 Blinding and sealing

This is the part of the protocol we care most about getting right, and it costs nothing to do.

	Action	Who
1	Freeze the model. After this point nobody touches a hyperparameter, a feature set or the training data	Computational pair
2	Write the predicted ranking of the three cell lines for each agent, plus the predicted statin pairing, into one document	Computational pair

3	Date it, sign it, hand it to the supervisor, and keep a second copy somewhere with a timestamp	Supervisor
4	Relabel the agents A to H. The bench team gets a code, a vehicle and a concentration range, and nothing else	Supervisor
5	Run and analyse every assay under the codes. Lock the analysis before anything is unblinded	Bench team
6	Break the codes and compare the prediction against the measurement	Everyone

6.3 The cells

Cell line	What it is	Why it is in the panel	Status
MDA-MB-231	TNBC, mesenchymal-like	The main model. Erastin IC50 near 2.2 uM and ferrostatin-1 reversible death are already published for this line [10], so it is the arm most likely to work	Core
MCF-7	Luminal, ER-positive	The comparator. It should be more resistant, and we can say why before we run it [9]	Core
MDA-MB-468	TNBC, basal-like	A second TNBC line, so the result is not about one line's quirks, and a third point for the ranking	Core
MCF-10A	Normal-like breast epithelium	Gives a selectivity index, which turns a killing result into a therapeutic window. Nice to have, not needed for the main claim	Only if the budget stretches

Conditions are identical across all lines and reported in full, because System xc- inhibitors are sensitive to how much cystine and glutamate the medium contains and to which serum lot you used. One medium formulation, 10% FBS from a single lot, 1% penicillin and streptomycin, 37 C and 5% CO2. Passaged below 80% confluence, used between passage 5 and 20, mycoplasma-screened before the campaign starts.

6.4 The drugs

The ranges below are starting brackets taken from published values. Each one gets narrowed in a pilot before the real runs. Vehicle is capped at 0.1% DMSO and every plate carries a vehicle-matched control.

Role	Agent	What it does	Starting range	Cost and supply
Type I inducer, our workhorse	Sulfasalazine	Blocks System xc-, starves the cell of cystine, glutathione falls, GPX4 stops working [9]	50 to 1000 uM, 8 points, 2-fold	Very cheap and registered in Egypt. This is the affordability argument of the whole project
Type I inducer, reference	Erastin	Same target, and the compound everyone else in the field uses. It is metabolically unstable in animals, which does not matter in a dish [11]	0.1 to 20 uM, published IC50 near 2.2 uM [10]	Moderate, research-grade import
Type II inducer	RSL3	Binds and inactivates GPX4 directly [3]	0.01 to 5 uM	Expensive. Budgeted for one cell line only, and dropped without harm to the main claim if the quotation comes back too high
Sensitiser,	Simvastatin	Blocks HMG-CoA reductase, so	Fixed sub-lethal	Cheap, registered

combination only	or lovastatin	IPP falls, the selenocysteine tRNA is not modified and GPX4 synthesis suffers [18]	dose near IC20, set in a pilot	generic in Egypt
Rescue control	Ferrostatin-1	Traps lipid radicals. If it rescues, the death was ferroptosis [1]	1 to 2 uM alongside the inducer	Moderate, and not optional. Without it we cannot make the central claim at all
Rescue control	Deferoxamine	Chelates iron. If it rescues, the death needed iron	50 to 100 uM alongside	Cheap and registered in Egypt
Rescue control	N-acetylcysteine	Puts cysteine back. Should rescue the Type I inducers and not RSL3, which is a clean way to tell the two mechanisms apart	1 to 5 mM alongside	Very cheap
Specificity control	Z-VAD-FMK	Pan-caspase inhibitor. If it fails to rescue, the death was not apoptosis	20 to 50 uM alongside	Moderate. This is the control that closes the argument

6.5 Everything we need to buy

Category	Items
Cells	MDA-MB-231, MDA-MB-468, MCF-7, and MCF-10A if funds allow. Authenticated stocks, mycoplasma-screened
Media and supplements	Base medium in one formulation for all lines, heat-inactivated FBS from a single lot, penicillin and streptomycin, L-glutamine, trypsin-EDTA, sterile PBS, HEPES, sodium bicarbonate, culture-grade DMSO, freezing medium
Test agents	Sulfasalazine, erastin, RSL3, simvastatin or lovastatin
Controls	Ferrostatin-1, deferoxamine mesylate, N-acetylcysteine, Z-VAD-FMK
Viability	MTT reagent and solvent, crystal violet, fixative, acetic acid for elution, LDH release kit
Redox and lipid peroxidation	DCFH-DA for general ROS, MDA or TBARS kit, GSH and GSSG kit, BODIPY 581/591 C11 if a flow cytometer turns out to be available
Imaging	Fluorescein diacetate, propidium iodide, DAPI or Hoechst 33342
Plastics	96-well plates in clear and black-walled, 6-well plates, T-25 and T-75 flasks, serological pipettes, filter tips, 15 and 50 mL tubes, 0.22 um syringe filters, cryovials, reservoirs, microcentrifuge tubes
General	Gloves, 70% ethanol, disinfectant, biohazard bags, parafilm, foil, sterile water
Equipment we already have	Class II cabinet, CO2 incubator, inverted phase-contrast microscope, fluorescence microscope, plate reader for absorbance and fluorescence, benchtop centrifuge, freezers, cell counter

6.6 What we actually run at the bench

We have split this into a core that has to happen and a set of extras we will run if time and money allow. Undergraduate projects fail more often from over-scoping than from bad ideas, so the core is deliberately small enough that three students can finish it properly in four months.

	Arm	What we do	What it answers
Core 1	Dose-	Sulfasalazine and erastin, 8 concentrations, all	Which line is sensitive and by how

	response	three lines, 24 h, n = 3. MTT with crystal violet as a second opinion. Six curves in total	much, which is the whole test of the prediction
Core 2	Rescue	The most sensitive line and agent pair at its IC50, with and without ferrostatin-1, deferoxamine, N-acetylcysteine and Z-VAD-FMK	Is this ferroptosis, or just cell death
Core 3	Combination	A sub-lethal statin dose plus a full inducer dose-response, on the line the model predicted would be resistant	Can we make a resistant line responsive on purpose
Core 4	Mechanism	The lead pair at its IC50 over 2, 6, 12 and 24 h. GSH and GSSG ratio, MDA or TBARS	Is glutathione falling and are lipids being peroxidised, as the mechanism requires
Extra 1	RSL3	Same dose-response format, MDA-MB-231 only	Does a direct GPX4 inhibitor agree with the Type I result
Extra 2	48 h timepoint	Repeat Core 1 at 48 h for the lead pair only	Does the ranking hold when the exposure is longer
Extra 3	Selectivity	MCF-10A alongside the tumour lines	Is there a therapeutic window

Every arm runs at n = 3 independent biological replicates with technical triplicates inside each, and every plate carries its own vehicle control. Core 1 is 18 plate runs. That is the number the timeline and the budget were built around.

6.7 The assays

Assay	How	What we read
MTT	Seed 5,000 to 10,000 cells per well, let them attach overnight, treat, add MTT at 0.5 mg/mL for 3 to 4 h, dissolve the formazan, read 570 nm against a 630 nm reference	Percent viability against vehicle, then IC50 from a four-parameter fit
Crystal violet	Fix, stain with 0.5% crystal violet, wash, elute in 33% acetic acid, read 570 nm	Adherent biomass, which catches the case where MTT is reporting a metabolic change rather than death
LDH release	Assay the supernatant against spontaneous and maximum-release controls	Percent cytotoxicity, independent of the viability readouts
FDA, PI and DAPI imaging	Stain live cultures, image at least three fields per well, quantify in ImageJ	Percent live, percent dead, and what the nuclei look like
DCFH-DA	Load the probe, treat, read fluorescence around 485 nm excitation and 530 nm emission	General intracellular ROS
MDA or TBARS	Lyse, react with thiobarbituric acid, read absorbance	Lipid peroxidation end-product, normalised to protein
GSH and GSSG	Lyse in kit buffer, assay both forms	The ratio, which is the most direct test of the Type I mechanism we can run on a plate reader
BODIPY 581/591 C11	Load, treat, measure the shift from red to green emission	Lipid peroxidation, the field standard. Only if we can get access to a flow cytometer

6.8 Statistics

- n = 3 independent biological replicates per arm, each in technical triplicate. That is the accepted minimum for cell-based dose-response and it is what the budget covers.
- IC50 from a four-parameter logistic fit on log concentration, reported with a 95% confidence interval rather than as a bare number.
- Comparisons between lines and treatments by two-way ANOVA with Tukey correction. Comparisons against a single control by one-way ANOVA with Dunnett. Alpha of 0.05.
- Rescue arms are analysed as the interaction between inducer and rescue agent, because that is the correct test for a rescue and a simple pairwise comparison is not.
- The model is scored by Spearman rank correlation between what it predicted and what we measured, with a bootstrap confidence interval. With three cell lines that interval will be wide, and we will report it wide rather than quietly leaving it out.
- Raw data, analysis scripts and figure code all go in the repository, so every number in the final report can be traced back to source.

7. What we expect to find

It is worth saying in advance what each result would mean, partly because it is good practice and partly because it forced us to check that no outcome leaves us with nothing to write.

If this happens	It means	Do we still have a project?
The predicted ranking matches the measured IC50 values, and ferrostatin-1 rescues	The main hypothesis holds. Expression-based prediction of measured ferroptosis sensitivity works, and we can show it was not fitted after the fact	Yes, and this is the best version
The ranking is only partly right, but TNBC still comes out more sensitive than MCF-7	The subtype-level biology is real and the line-level resolution is not there yet. We say which is which	Yes. An honest calibration is publishable
The prediction fails, and Aim 0 shows the circular model would have claimed high accuracy on the same data	We have measured the inflation problem directly instead of asserting it	Yes, and this may be the most useful thing we produce
The statin combination shifts the IC50 significantly	We designed a sensitisation and it worked, which is a stronger claim than predicting a sensitivity that was already there	Yes, and it is the result with the clearest translational hook
Ferrostatin-1 does not rescue	Whatever we are seeing is not ferroptosis. We report that, and use the LDH and caspase readouts already in the design to say what it might be instead	Yes. The rescue panel is there precisely to catch this

No row in that table ends with the students having nothing to submit. That was a design constraint rather than an accident, and it is the main reason we think this is a sensible project for a final undergraduate year instead of an ambitious one that quietly collapses in month seven.

8. What we will hand in

Deliverable	Target	Month
Harmonised dataset	Expression joined to measured response, with per-agent sample sizes stated	2
Aim 0 comparison	Both labelling schemes on identical splits, gap reported	3
Model performance	Held-out Spearman with a confidence interval, or an honest negative result	4
Sealed prediction	Dated and signed before the first plate	5
Core bench campaign	3 lines, 2 inducers, 8-point curves, n = 3, plus rescue, combination and mechanism	8
Ferroptosis confirmed	At least one agent and line pair rescued by ferrostatin-1 at p below 0.05, and not rescued by Z-VAD-FMK	8
Measured model accuracy	Predicted ranking against measured IC50, interval included	9
Web tool	Deployed, documented, tried by at least three people outside the team	9
Cost analysis	Cost per treatment course at Egyptian prices for every agent that worked	9
Written outputs	One poster, one manuscript submitted, one competition entry	9

9. Timeline

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9
WP1 data engineering	X	X							
Coding ramp-up	X								
WP2 modelling and Aim 0		X	X	X					
WP3 lock, seal, order, revive cells				X	X				
Core 1 dose-response					X	X			
Core 2 and 3, rescue and combination						X	X		
Core 4 mechanism							X	X	
Extras, if time allows							X	X	
WP5 unblind, analyse, deploy								X	X
Writing, poster, competitions								X	X

10. Budget

Total requested is 100,000 EGP, which is 20,000 EGP for each of the five students.

	Item	Why we need it	EGP
1	Cell lines MDA-MB-231, MDA-MB-468 and MCF-7, plus revival, freezing and mycoplasma screening	Two TNBC lines and the ER-positive comparator. Bought once, used in every arm	13,500
2	Media, serum, antibiotics, trypsin, PBS, DMSO	Four months of continuous culture on three lines, around 35 plate runs including the extras	21,000

3	Sulfasalazine, erastin and RSL3	The compounds under test. Sulfasalazine is the cheap workhorse, RSL3 the expensive reference we can drop	12,000
4	Statin	The combination arm	1,500
5	Ferrostatin-1, deferoxamine, N-acetylcysteine and Z-VAD-FMK	The controls that decide whether we are allowed to use the word ferroptosis at all	13,000
6	MTT, crystal violet, LDH kit, DCFH-DA, FDA, PI, DAPI, TBARS kit, GSH and GSSG kit	Every readout in the validation and the mechanism arms	18,000
7	96-well and 6-well plates, flasks, filter tips, pipettes, syringe filters, cryovials	Three lines by two inducers by eight points by three replicates, plus the rescue and mechanism arms	11,500
8	Contracted assay service, held in reserve	Protects the bench deliverable if facility access is interrupted. Released only if needed	3,000
9	Compute, storage and hosting for the web tool for 12 months	Runs the pipeline and keeps the tool online after we graduate	2,000
10	Poster printing, conference registration, publication fee contribution	The dissemination outputs	4,500
	Total		100,000

These figures show the shape of the spend and the proportions between lines. Replace every one of them with a real supplier quotation before submitting, keep the total at exactly 20,000 EGP times the number of students, and try not to leave every line as a round number. Itemised quotes read as a plan. Round numbers read as a guess.

11. Who does what

Person	Job	Why they can do it
Student 1, computational	Pulling, harmonising and checking the data (WP1)	Data handling coursework, and the ETL pipeline is shared with student 2
Student 2, computational	Modelling, the Aim 0 baseline, interpretation and the web tool (WP2 and WP5)	Wrote the Python and scikit-learn pipeline this proposal is built on
Student 3, bench	Culture maintenance, MTT and crystal violet (WP4)	Departmental practical training in cell culture
Student 4, bench	Fluorescence imaging and image quantification (WP4)	Trained on the imaging protocols already running in the supervising laboratory
Student 5, bench	Redox assays, the rescue arms and the statistics (WP4 and WP5)	Trained on the assay protocols already running in the supervising laboratory
Supervisor	Study design, custody of the sealed prediction, blinding, quality control	Runs a laboratory where MTT, LDH, crystal violet, FDA with PI and DAPI, DCFH-DA, JC-1 and caspase-3 assays are already validated and in routine use

Worth saying plainly to whoever reviews this. Not one method in the bench half is new to the laboratory we are working in. All of them are running there now. The risk in this project is not whether we can perform the assays. It is whether the model is any good, and the design is built so that we find out either way.

12. After the project

We are being careful not to oversell this part. A nine-month undergraduate project does not produce a company. It can produce two things that outlast it, plus a realistic idea of what would come next.

Question	Answer
What is left behind?	A web tool that scores an expression profile for ferroptosis vulnerability and returns a ranked list of affordable inducers, and a drug combination with in-vitro evidence behind it in TNBC models
Who would use it?	Laboratories doing molecular profiling, groups running repurposing screens, and academic labs who want a prioritisation step before committing bench time. The tool is aimed at researchers first, not at clinics
Why would they use ours?	Because it reports how often it was right against measured killing, which the published signatures cannot do, and because the drugs it points at are cheap
What would it cost a patient?	Sulfasalazine and simvastatin are registered generics in Egypt. We will publish the cost per course at local prices, which is the number that matters in a system where drug price limits access
What is the honest next step?	Extending the model to another high-burden cancer, and a retrospective look at Egyptian tumour samples with an oncology partner. Both need a follow-on grant, and neither is promised here
Who we would want to work with	The Academy of Scientific Research and Technology, the Innovators Support Fund, a contract research provider for the assays we cannot run ourselves, and the university technology transfer office

Getting the work out

Where	What	When
Departmental seminar	Interim results, then final results	Month 5 and month 9
Conference	Poster at a national biochemistry, oncology or bioinformatics meeting	Month 9
Journal	Manuscript submitted, with the Aim 0 comparison as a separate contribution	Month 9
Public repository	Pipeline, sealed prediction document and analysis notebooks	Month 9
Web	The FerroSelect tool, deployed with a short user guide	Month 9
Competitions	National student research and innovation competitions	Months 8 and 9

13. Ethics and safety

- No human subjects, no animals, no identifiable patient data. We use immortalised cell lines and de-identified public datasets, so institutional review board approval is not required. We will still notify the department, and approval becomes necessary if the follow-on plan to look at Egyptian tumour samples goes ahead.
- Every data source is public and free: DepMap and CCLE, GDSC, CTRPv2, PRISM, TCGA and METABRIC through cBioPortal, GEO and FerrDb. Academic reuse is permitted, and we cite and respect the conditions of each.
- Analysis code is released open-source, and the sealed prediction is timestamped so that anyone can audit the claim that the prediction came first.

- Cytotoxic agents, iron chelators and organic solvents are handled in a certified cabinet with the usual protective equipment and a separate chemical waste stream.
- Cell lines stay inside a defined passage window, with mycoplasma screening documented and provenance recorded for every stock.

14. References

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Check every one of these against the original paper and reformat to whatever style the funder or journal asks for before you submit.

Annex A. Plate layout

One plate per cell line per agent. Columns 1 and 12 hold sterile PBS and are left out of the analysis, because edge wells evaporate differently from the rest of the plate. The eight concentrations run across columns 3 to 10 as a 2-fold series, in technical triplicate down the rows, with vehicle and blank controls on the same plate rather than on a separate one.

	1	2	3	4	5	6	7	8	9	10	11	12
A	PBS	Blank	C1	C2	C3	C4	C5	C6	C7	C8	Vehicle	PBS
B	PBS	Blank	C1	C2	C3	C4	C5	C6	C7	C8	Vehicle	PBS
C	PBS	Blank	C1	C2	C3	C4	C5	C6	C7	C8	Vehicle	PBS
D	PBS	Untr.	C1	C2	C3	C4	C5	C6	C7	C8	Vehicle	PBS
E	PBS	Untr.	R1	R2	R3	R4	R5	R6	R7	R8	Vehicle	PBS
F	PBS	Untr.	R1	R2	R3	R4	R5	R6	R7	R8	Vehicle	PBS
G	PBS	Blank	R1	R2	R3	R4	R5	R6	R7	R8	Vehicle	PBS
H	PBS	Blank	Fer-1	DFO	NAC	ZVAD	Fer-1	DFO	NAC	ZVAD	Vehicle	PBS

C1 to C8 is the inducer alone. R1 to R8 is the same series with the fixed sub-lethal statin dose, which is the combination arm. Row H holds the rescue agents at the inducer IC50, four of them with inducer and four without. Blank is medium with no cells. Untr. is cells with no treatment.

Annex B. Which sections go to which funder

Scheme	Amount	Sections to send	What to change
ASRT graduation projects	20,000 EGP per student, capped at 100,000	1, 2, 4, 6, 8, 9, 10, 11, 12, 13	An Arabic title and a 200-word Arabic summary are compulsory. Use the companion file of Arabic answers, and remember to drop the copper wording carried over from the old draft
ISF entrepreneurial graduation project	Up to 100,000 EGP	1, 3, 4, 6, 8, 11, 12, 13	Expand section 13 into a one-page business model canvas. This scheme weights commercial potential heavily, so keep the honest tone but give the product half more detail
InnovEgypt and TIEC	Training and accelerator rather than a research grant	3 and 13	Send the commercialisation half only. Treat it as a credential and a way to build the business case
University internal competitions	Varies	1 to 12	Usually accepted as it stands

Do not send the same budget lines to two funders. If two applications succeed, split the work packages so that one scheme pays for the bench campaign and the other pays for the tool and the dissemination, and tell both of them you did.